

Bertrand's Paradox & Invariant Measure Selection

Analytical Derivations of Geometric Probability under Differing Symmetry Models

A Rigorous Analytical Derivation, Stochastic Analysis, and Empirical Verification

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Abstract

This paper presents a complete analytical evaluation of Bertrand's Paradox in geometric probability.

By formulating chord selection over a unit circle under three distinct physical definitions of uniform randomness—Random Endpoints ($P = 1/3$), Random Radius Offset ($P = 1/2$), and Random Midpoints ($P = 1/4$)—we demonstrate how continuous geometric state spaces depend on the chosen symmetry measure.

We derive the analytical probability for each interpretation, evaluate Jaynes' Principle of Maximum Ignorance as a physical invariance heuristic under continuous transformation groups, and evaluate a companion interactive web simulation.

Live Simulation Engine: <https://bertrandsparadox.vercel.app/>

Core Math Domain: Geometric Probability & Measure Theory

Primary Focus: Invariant Measures and Geometric State Space Parameterization

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1 Problem Formulation & Physical Geometry

Consider a circle of radius R circumscribing an equilateral triangle with side length $L = \sqrt{3}R$. A chord is selected "at random" across the circle.

The classical paradox formulated by Joseph Bertrand in 1889 asks for the probability P that the length C of the random chord strictly exceeds L .

$$P(C > \sqrt{3}R) = ? \tag{1}$$

Equation (1) establishes the primary inquiry. Because "randomly chosen" is ill-defined over continuous state spaces without specifying a probability space $(\Omega, \mathcal{F}, P)$, different choices of parameterization yield distinct, mathematically valid probability measures.

1.1 Primary Investigation Objectives

The core analytical goals of this research paper are defined across two operational domains:

- **Analytical Derivation:** Formulate and derive the exact closed-form probabilities for the three classical interpretations ($P_1 = 1/3$, $P_2 = 1/2$, and $P_3 = 1/4$).
- **Invariant Measure Analysis:** Apply E.T. Jaynes' Transformation Group Method as a physical symmetry heuristic to evaluate rotational, translational, and scale invariance.

2 Theoretical Proofs & Classical Interpretations

2.1 Method 1: Random Endpoints (Uniform Angle Parameterization)

Fix an arbitrary point A on the boundary S^1 . Select a second endpoint B uniformly along the circumference, parameterized by angle $\theta \sim U(0, 2\pi)$.

An inscribed equilateral triangle with vertex at A divides the circumference into three equal arcs of angle $2\pi/3$.

$$P_1 = \frac{\text{Favorable Arc Length}}{\text{Total Circumference}} = \frac{2\pi/3}{2\pi} = \frac{1}{3} \quad (2)$$

Equation (2) establishes the outcome under endpoint parameterization. The chord exceeds $\sqrt{3}R$ if B falls within the arc subtended opposite A , yielding $P_1 = 1/3$.

2.2 Method 2: Random Radius / Offset (Uniform Distance Parameterization)

By rotational symmetry, fix the direction of a radius vector of length R . Select a point D uniformly along this radius at distance $r \sim U(0, R)$, and construct a chord perpendicular to the radius at D .

By Pythagorean geometry, the chord length $C(r)$ as a function of radial offset r is derived directly.

$$C(r) = 2\sqrt{R^2 - r^2} \quad (3)$$

Equation (3) gives chord length $C(r)$. Setting $C(r) > \sqrt{3}R$ determines the valid offset boundary.

$$2\sqrt{R^2 - r^2} > \sqrt{3}R \implies 4(R^2 - r^2) > 3R^2 \implies r < \frac{R}{2} \quad (4)$$

Equation (4) proves that the chord length exceeds the triangle side if and only if $r \in [0, R/2)$.

$$P_2 = \frac{R/2}{R} = \frac{1}{2} \quad (5)$$

Equation (5) calculates the ratio of favorable radius length to total radius, yielding $P_2 = 1/2$.

2.3 Method 3: Random Midpoint (Uniform Planar Area Parameterization)

Select a point (x, y) uniformly over the interior 2D disk area $\Omega = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq R^2\}$. This point serves as the unique midpoint of the chord.

A chord's length exceeds $\sqrt{3}R$ if and only if its midpoint lies strictly inside a concentric circle of radius $R/2$.

$$P_3 = \frac{\text{Area(Concentric Disk)}}{\text{Area(Total Circle)}} = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4} \quad (6)$$

Equation (6) evaluates the area ratio measure, yielding $P_3 = 1/4$.

3 Jaynes' Invariance Principle & Physical Resolution

In 1973, Edwin T. Jaynes demonstrated that when modeling physical experiments (such as tossing long thin straws onto a circular target), a physical setup should ideally be invariant under scale, rotation, and translation transformations.

3.1 Scale and Translation Invariance Heuristic

Let $f(r, \theta)$ be the probability density of chord midpoints in polar coordinates. Requesting independence from circle radius R enforces a functional transformation under continuous translation and scaling groups.

$$f(r, \theta) = \frac{1}{2\pi r R} \tag{7}$$

Equation (7) represents the probability density that satisfies continuous spatial translation and scale invariance, matching Method 2 ($P_2 = 1/2$). It is important to emphasize that Jaynes' result represents a physical invariance criterion for toss experiments rather than a measure-theoretic proof that invalidates $P_1 = 1/3$ or $P_3 = 1/4$.

- **Random Endpoints** ($P_1 = 1/3$): Assumes uniform rotational density on the boundary S^1 .
- **Random Radius Offset** ($P_2 = 1/2$): Satisfies maximum ignorance under continuous physical translation/scale symmetry (Jaynes' Principle).
- **Random Midpoints** ($P_3 = 1/4$): Assumes uniform spatial density over the interior 2D disk measure.

4 Applications and Practical Observations

Bertrand's Paradox illustrates how continuous parameter mappings require explicit probability space specification across applied domains:

4.1 Continuous Parameter Mapping

In financial modeling and spatial risk analysis, switching parameterizations or coordinate systems requires careful specification of the underlying probability space $(\Omega, \mathcal{F}, P)$ to avoid introducing unintended density distortions.

4.2 Spatial Risk & Network Percolation

In spatial point process models and high-frequency Wireless Sensor Networks (WSNs), calculating link coverage probabilities depends heavily on whether edge lengths are sampled uniformly by angle, distance, or area.

5 Interactive Web Simulation Evaluation

To complement this theoretical monograph, an interactive single-file web application was developed: `bertrand_paradox_sim.html`.

- **Live Interactive Application:** <https://bertrandsparadox.vercel.app/>
- **GitHub Source Code Repository:** <https://github.com/wheyyydant/math-drafts-and-si>
- **Real-Time Sampling Mode Switcher:** Allows users to toggle dynamically between Method 1 (Endpoints), Method 2 (Radius Offset), and Method 3 (Midpoint Area) to observe convergence toward $1/3$, $1/2$, and $1/4$ respectively.
- **HTML5 Vector Canvas Field:** Displays thousands of stochastic chords drawn over the circumscribed unit circle, visually highlighting long chords ($C > \sqrt{3}R$) in real time.

6 Concluding Remarks

We have presented an analytical derivation of Bertrand's Paradox, demonstrating that probabilities over continuous parameter spaces are well-defined only after specifying a symmetry transformation group and probability measure.

7 Mathematical Glossary

- Probability Space:** A mathematical triple $(\Omega, \mathcal{F}, P)$ defining a sample space, σ -algebra of events, and probability measure.
- Scale Invariance:** The property where a probability distribution's structure remains unchanged when spatial dimensions are rescaled by a factor k .
- Jaynes' Principle:** A heuristic specifying that when an outcome's physical problem statement does not specify initial conditions, the density should be invariant under transformation groups representing complete physical ignorance.
- Uniform Density:** A distribution where equal volumes or arc lengths in parameter space carry equal occurrence probabilities.